



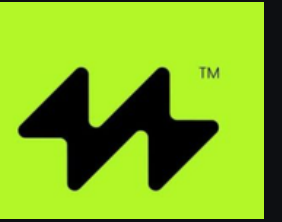
The lights haven't gone out yet.

VOLTRA sees that they are going to.

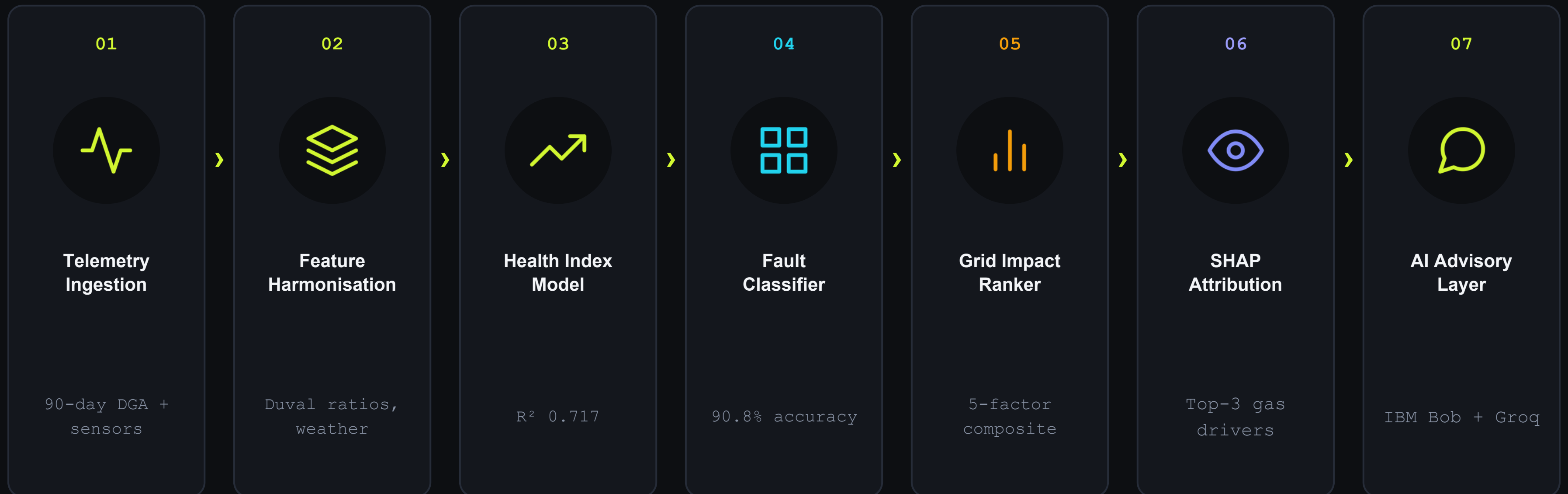
Predictive Grid Intelligence for the Anand District Transformer Fleet — real-time risk scoring across 18 assets, 545 MVA, 4 sub-zones.

FLEET AT A GLANCE

18 Transformers Monitored 4 Grid Sub-Zones	545 MVA Fleet Capacity 11kV - 132kV	90.8% Fault Classification Acc. 7 IEC fault classes	+89d RUL Recovered · TX-115 ₹4.2 Cr avoided
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From dissolved gas to dispatch order — one automated pipeline.



ISOLATION GUARANTEE: Stages 1-6 run on pure Python + scikit-learn. Every LLM call is wrapped in try/except — scoring never breaks.



Two trained models. Zero simulated metrics.

Both trained on real Kaggle transformer datasets, cross-validated, limitations documented.



MODEL 1 · Health Index Regressor

0.717

R² Score

5.88 HI

MAE

470

Training rows



MODEL 2 · DGA Fault Classifier

90.8%

Accuracy

0.896

F1 Score

4,151

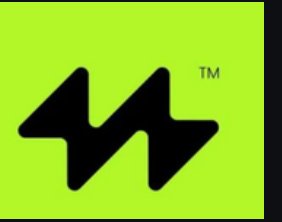
Training rows

FAULT CLASS SEVERITY · RANKED BY RISK SCORE



RESPONSE URGENCY

D2	High-energy arcing	24 hrs
T3	Severe oil overhear	48 hrs
D1	Intermittent sparking	7 days
T2 / T1 / PD	Thermal & discharge	14-30 days



Every screen built for one person: the grid operating engineer.



Live Grid Inspector

18 assets ranked by live risk score, color-coded tiers.

`/grid`



7-Day Maintenance Plan

Auto-generated crew, asset & action calendar.

`/plan`



What-If Simulator

Load & weather stress-test, scores recalc live.

`/simulate`



Live Transformer Map

18 pins on real coordinates, synced to the grid.

`/map`



Prediction Studio

Upload any CSV, get ranked risk instantly.

`/predict`



One intervention. **+89 days** of life recovered.

**ACT 1****Degradation · Days 60–78**

Acetylene surged to 1,420 ppm. D2 arcing confirmed.
RUL fell to 7.7 days.

**ACT 2****VOLTRA Triggers · Days 78–79**

Risk score hit 0.94. Bob dispatched a crew within 16 hours.

**ACT 3****Recovery · Days 80–89**

C₂H₂ dropped to 48 ppm. RUL recovered to 97 days.
Service restored.

RUL:

7.7 DAYS**97 DAYS****+89.4d**

Net life recovered

**SMS**

Live SMS integration

**38,000+**

Households protected



A score you can defend to a regulator.

An advisory you can act on immediately.



SHAP Explainability

- HI = 67.3 decomposed: C₂H₂ +12.4, CH₄ +9.1, H₂ +5.6
- Bob gets top-3 SHAP features with every score
- Operators present attribution to CERC as evidence



Prompt-Injection Defense

- Field reports regex-filtered before reaching any LLM
- Rejected inputs logged for forensic review
- A report can never override sensor data — capped 1.25×

LLM ISOLATION: Core scoring runs on pure Python + scikit-learn — zero API dependency. Every response tags its `advisory_source` (real LLM vs deterministic fallback).

4 AI SERVICES — EACH USED FOR WHAT IT DOES BEST



IBM Bob

Claude 3.5 Haiku · Advisory



Groq LPU

Llama-3.3-70B · Reports <2s



Gemini 3.6

Geospatial hazard search



Open-Meteo

Live weather · no key



Built end-to-end in 48 hours. **Grounded in 4,621 real rows.**



ML Pipeline

scikit-learn · SHAP



Backend

FastAPI · Pydantic v2



Frontend

React 19 · TanStack



Mapping

MapLibre GL



AI Layer

Claude · Groq · Gemini

TEAM TECHTONICS

● Om Rashiya

Lead · Backend + Frontend + Map

● Yash Bharvada

Python ML Models + API

● Nikunj Desai

Python ML Models + API

● Purva Shah

Architecture + Testing + SMS configuration

GROUNDING PREDICTIONS. BEFORE THE DARK.

VOLTRA turns a transformer's last 90 days of gas data into a maintenance calendar, a crew dispatch order, and a plain-English advisory.

[/grid](#)

[/map](#)

[/predict](#)

[/technology](#)

FastAPI :8000

CHARUSAT · ANAND